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Enhancing the Security of Cloud Computing: Genetic Algorithm and QR Code Approach

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***Abstract*— Paradigms of Cloud computing are gaining prevalent acceptance due to the numerous benefits they provide. These comprise cost-effectiveness, time savings and effective deployment of computing resources. However, privacy and security issues are among the major hindrances holding back the widespread adoption of this novel technology. In this context, enhancing the security of cloud computing demands the highest priority based on the popular encryption method RSA, Genetic Algorithm (GA), and 3D Quick Response (QR) code using MAC address. This paper will show an approach of amplifying the security of cloud computing. Therefore, we are proposing advanced two-step security mechanism of cloud computing. Firstly, RSA is used to provide initial protection of the first stratum. RSA approaches message as input to the hash function and generated hash values is encrypted by using generated sender key. And the receiver will generate a hash code and decrypts the message using the sender's public key. Finally, The Genetic algorithms (GA) are generalized search algorithms based on the mechanics of natural genetics. GA maintains a population of individuals that represent the candidate solutions. After successful completion of Genesis, mutation and crossover GA will provide a framework to match the desired pattern. Moreover, a media access control address (MAC address) of a device is a distinctive identifier assigned to network interfaces for communications at the data link layer of a network segment. In this context, the MAC address is used to generate the 3D Quick response code for user verification. Through experiments, we affirm that the combination of GA and 3D QR Code approach with RSA for two-step security mechanism shows strong potential to heighten the security of Cloud Computing.**

Keywords—cloud computing; RSA algorithm; genetic algorithm;Face Recognision; 3D QR code;

I. INTRODUCTION

Cloud computing is more intimate in this innovative epoch. This is a general meaning for used to compromise a new step of network-based computing that can take place over the Ethernet. It is a great attention from individuals at home to all governments in both of publications and among users. It is clear to state, Cloud computing becomes more popular to store data and using data in the network without using hardware. Now a day the security statement is a big question of Cloud Computing. There are the most important data and the information is losing in every day for the lack of ensuring security. The Security enhancement is the lone means to reach the end. This enhancement can be processed by RSA algorithm, Face Recognition by Genetic Algorithm, Generate 3D QR code of a unique MAC address. RSA cryptosystem was designed in 1977 by Ron Rivest, Adi Shamir, and Leonard Adelman. This system is a more popular public key algorithm for its simplicity. This security stands to compute the elements of a big composite integer. Currently factoring 1024 bits integer is assumed to be as complex as the workload of 280 which is the current benchmark used in cryptography. The high management cost and storage usage bring the algorithm in the research area [1]. Facial expression recognition by a computer can be parted into two approaches are constitutionally based and facing-based. In constituent-based access, recognition is founded on the relationship in human facial characteristics like as eyes, lip, nose, profile silhouettes and face boundary [2]. Genetic algorithm, which is now a day at best used and efficient algorithm in the passing generation. It mainly performs three steps for checking. Those steps are Mutation, Crossover, and Genesis. The persistence of mutation in GAs is conserving and introducing diversity. It is a genetic operator, which maintains genetic diversity from one generation of a population of genetic algorithm chromosomes to the next. Then the crossover is the other constituent of the Genetic algorithm which mainly applied for the matching. In our operation, this is the main ingredient. Genesis is the concluding component of GA which is used for mutation and crossover formula in the programming section [3]. MAC is a lasting and very unique identity of a device with networking capability which is recognized also as a Network Interface Card (NIC). The MAC address is a 48-bit binary number that is contained in plaintext frame which is transmitted in network connectivity [4]. The MAC address is determined to exploit with the Data link layer. It also is known as the Physical address of a hardware. A barcode is an optical machine-readable representation of data, which presents information about attaching the object. 3D QR code is composed of an array 3D cells, called modules, it became either filled or empty, corresponding to possible values of a piece. The created 3D

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construction allows for much more data-storage capability inside the codification. It contains data not just in its x-axis and y-axis but in depth also [5].

II. Related Work

There exist some boundless research works by some prodigious authors regarding the security enhancement of cloud computing. Reference [1] shows that the RSA algorithm can ensure the password layer security effortlessly. Simply, by using some unethical approaches, the layer can escape nowadays. Another in reference [3] indicates that the Genetic Algorithm is applied as the detection of the human face with some edge of the facial. By doing mutation and crossover formula the method was well perfect for accuracy to make sure the security for face recognition technique. So the combination of these techniques will provide advanced security in layer two with facial recognition. The 3D QR code is also a security technique which is clearly followed out with reference [6]. Mac binding is too used as preferred login in the second layer for users. Hash technique is practiced where the registration of the device falls out. During the registration process, the MAC is encrypted as token and stored in the database. During the time of login user need to pass through a bar code reading mechanism. The scanned barcode is then transmitted to the server for a successful match with Hash System. Our suggested approach is to make the 2-layer security wherein the first layer RSA system will take place and in the second layer, there will be a preference for users to choose between facial recognition and 3D QR code with MAC binding process.

III. Methodology

In this part, we will discuss how our proposed scheme will function in real life. Methodology section consists of several subsections where encryption technique, registration process, and data matching techniques will be explained in details with their respective process diagram and flow chart. Fig.1. indicates the flow chart of RSA algorithm where the running operation of that particular algorithm shows in particular.

A. Username and Password Encryption and Decryption

Signing process of users will extend through the RSA algorithm where the encryption and decryption will take place. Two step procedure for user registration is clearly depicted in Fig.1. With a flowchart.

Step-1: The RSA system will operate both public and private key where it will construct a combination of these and encryption of messages will also be performed employing the same method.

Step-2: Here, the values are encrypted and decrypted with $C=M^E \bmod N$

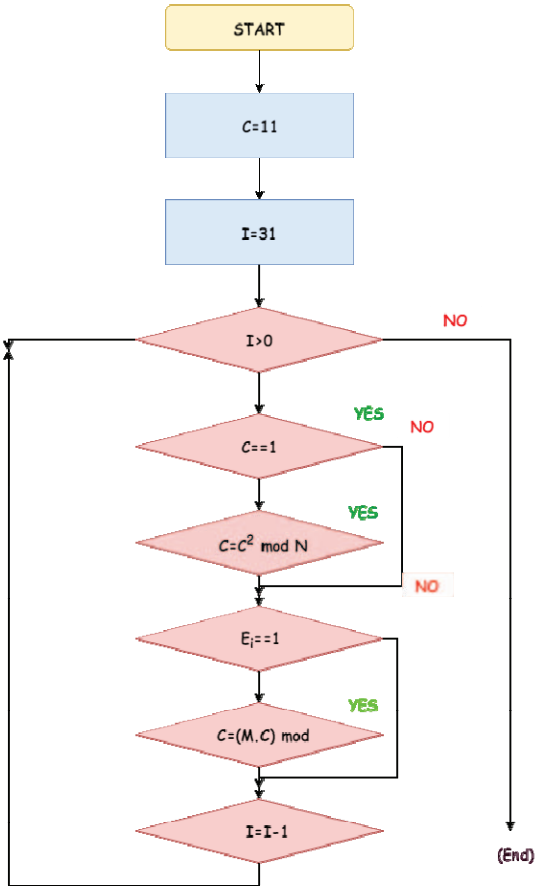

Fig. 1. Flowchart of RSA Algorithm for encryption and decryption

B. Registration Process

After successful completion of the initial steps, the following component of registration offers user 2 types of alternative selections such as Face Recognition and 3D QR code. Explanation and working process of each step are summarized here. Fig.2. Indicates that how this registration model works for users.

Step-1: In step 1 initial security layer of the developed system is shown where the database is used to complete the registration process. Users need to enter [username, password] for the login method. The username will be taken to the database with some garbage string and the password will be encrypted using RSA cryptosystem.

Step-2: After Encryption, this system sets up two options for users (Option 1 or Option 2)

Choice 1: The Genetic algorithm is applied to recognize the facial expressions, whereas in this composition, we will attempt 5 sample images to put through this technique.

Choice 2: By using the MAC address of device option 2 will generate a unique data for QR code.

Step-3: Execution of successful data matches with the database, so the user will receive the permission to access.

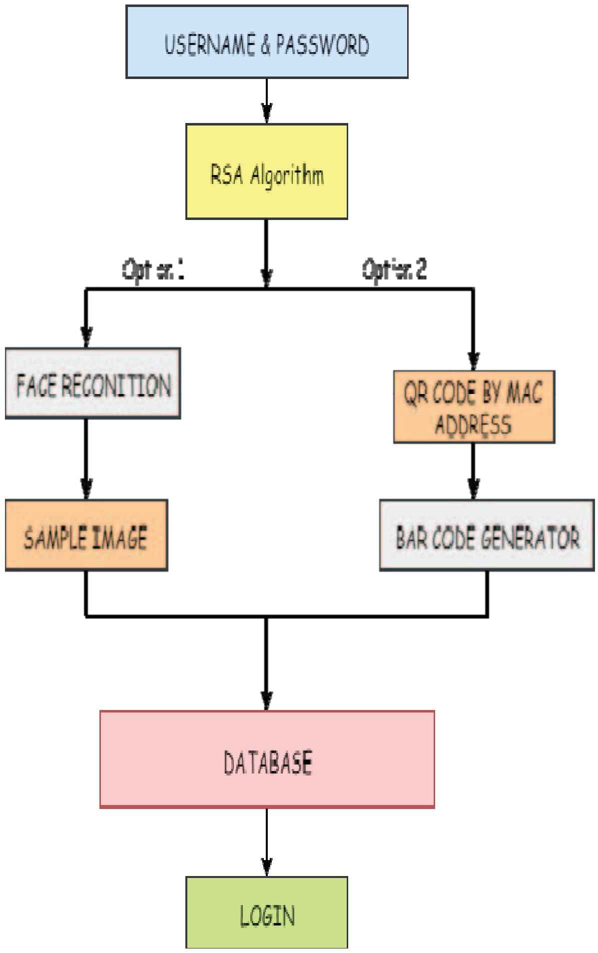

Fig. 2. Proposed model of user registration process

C. Data Matching

Among all data, matching is the most vital part of our model where our architecture needs to verify the input data by comparing with stored data from the database. In the contingent process of this technique is entirely designed in Fig.3. Stepwise procedure is narrated to a lower place

Step-1: The 1st step of data matching starts with username and password, which remain encrypted using the RSA algorithm. Encrypted data, then compared with the database. If the encrypted data accorded with the database, login gets the verification from the system and step 2 begins. If not matched, then it showed with "Login Error".

Step-2: After successful verification, the user will be able to choose the next step. It consists of two options where the user holds the privileges to select among these.

Option 1: This step works on users face recognition where the user's picture will be captured by the device HD camera. In this step there are 5 image samples will be taken from the user. Then Genetic Algorithm is applied to the device as a form of data for fitting with the database. If the information of the image is touched with the database, then the user gets the verified message to access the "Login" panel and if not, that's mean the user is not verified which indicates a "Login Error".

Option 2: This option will generate a QR code by utilizing the unique MAC address of the user's device. And it will be presented as a 3D barcode [8]. This barcode will be matched with the registered barcode which stored on the database. If the data of barcode match to the database, then the user will be able to access "Login" panel and if the data of the barcode is not matched, that's mean the user has not given the permission to access.

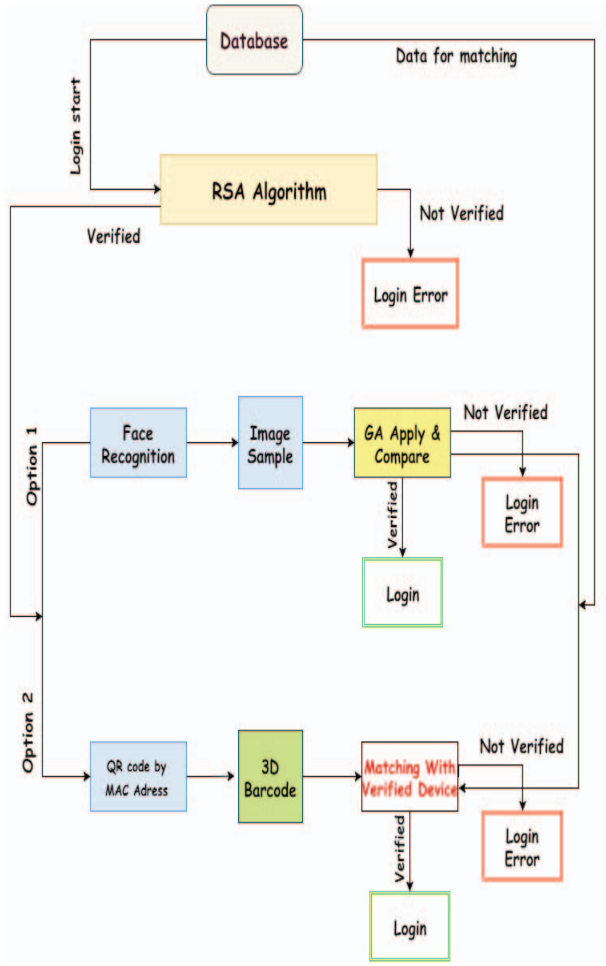

Fig. 3. Matching process of user data

IV. Implementation

In this part, we will establish how our suggested method will discover out the user by using three previously mentioned security technique. The implementation process will be shown for RSA, GA, and 3D QR Code Generator. Fig.4, Fig.5, & Fig.6, designates that how the implementation process is tested with real objects.

A. *RSA Algorithm*

Generally, RSA uses two exponents, p, and q, where p is public and q is privet key. Let the plaintext is M and C is ciphertext, then at encryption [7]. $C=M^p$ mod n and Decryption side become $M= C^q$ mod n. Hither, this method used to generate the 1st layer system to get this entire architecture. If user able to log in successfully, then the 2nd layer will be pictured, otherwise the error time will increase by showing an error message. The 2nd layer is based on user choice.

B. *Genetic Algorithm*

A genetic algorithm is considered as heuristics based search approach algorithm where chromosomes are treated as the main parameter. GA is used to explore and optimize the problems solutions. Basic operations of GA are reproduction, crossover, and mutation. For a successful experiment selection of the input image from train database is prerequisites where an individual selection of small pixel of that image is also obligatory. By applying crossover, it will able to taper out the amount of changes in the input test image comparison with train pictures. Up next for minimum matching the procedure will calculate ten generations of the image with matching and add up each amount with Boolean (0 and 1) separation. Eventually, those images whose sum value are minimized.

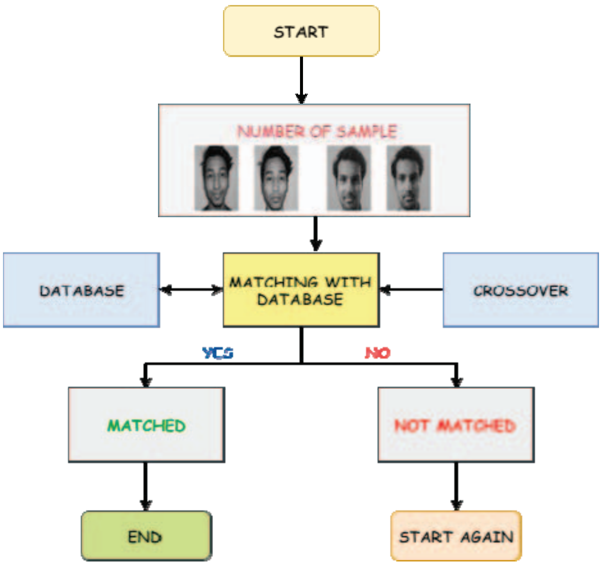

Fig. 4. A face matching procedure by using Genetic Algorithm

Genetic Algorithm consists basically of making a search in the image of the object that most resembles an ellipse of a proportional ratio of 1.5:1 [8].

The individuals, the ellipses are written using binary form, assuming that each individual has four chromosomes, respectively: position X, position Y, the ratio of X, and the ratio of Y.

The sample code is given bellow:

Position of	**X=**	**150**	**10010110**
Position of	**Y=**	**60**	**00111100**
Ratio of	**X=**	**95**	**01011111**
Ratio of	**Y=**	**80**	**01010000**

In MATLAB the iteration is conducted by the sample image matrix binary values. Fig.5 depicts the whole procedure of face recognition mechanism using GA with some stepwise operations. Following steps (1-4) demonstrate how MATLAB is used to verify the face of users by comparing existing images with the database. For the successful experiment, we have utilized a personal database to successfully spot a face in MATLAB.

Step-1: Implementing and running the codes of GA to add some sample images for testing.

Step-2: Bitmap (. bmp) image format is used in order to save samples for further use. Extract and upload process of samples are tracked by the database. These samples are employed to create a complete matrix for GA.

Step-3: Comparing two samples, namely *sample1.bmp* with *sample2.bmp*, the results show that mentioned samples are matched with each other. So the resolution designates the matching of experiment samples.

Step-4: Whenever the samples didn't match, results show the unsuccessful matching results.

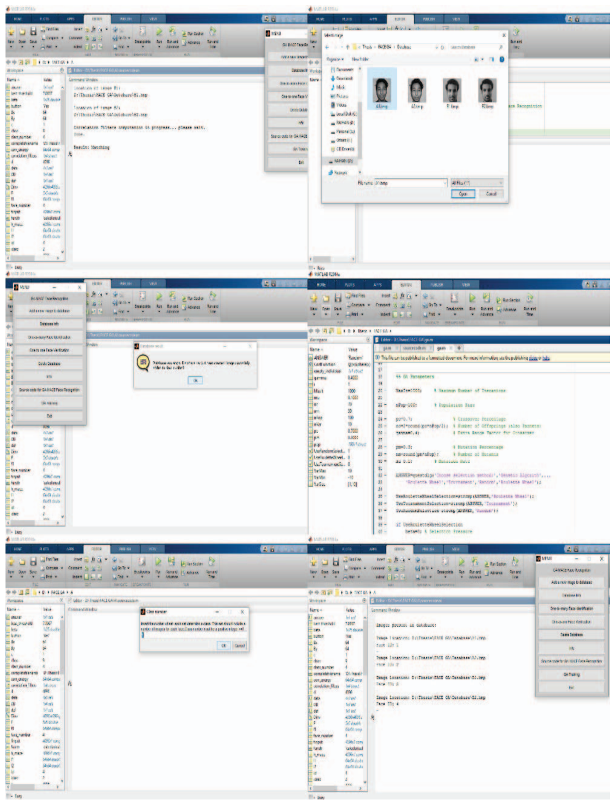

Fig. 5. Implementation of face recognition in MATLAB

C. *QR Code Generation*

Another portion of the registration process is based on QR code generation. Generation of QR codes depends on the MAC address of individual devices. QR codes are quickly arriving at high degrees of acceptance. More and more people adopt and use this technology every day. One of the reasons behind the rapid growth of the QR code is that it gains momentum as smartphone users grow across the world and marketers use QR codes to reach mobile consumers. [9] In this section, the device will need to register through a web page where that particular site will store the MAC of that device for the encoding purposes. Again in verification section proposed design system will shake off a barcode to registered users where the successfully registered device needs to scan that code through a scanner. If that code matched with the login, then the user will take the successful verification otherwise particular section will be sealed. A complete illustration of the above discussion is depicted in Fig. 6. For our experiment, a MAC address is applied to generate a QR code. Fig.6. Indicates the presence of MAC ID 78: F7:BE:B7:83:57 in that particular QR code. After successful scanning of that QR code we have found the existence of that exact MAC ID 78: F7:BE:B7:83:57 on that specific QR Code. Successful execution of the QR code generation for users is clearly exhibited in Fig. 6.

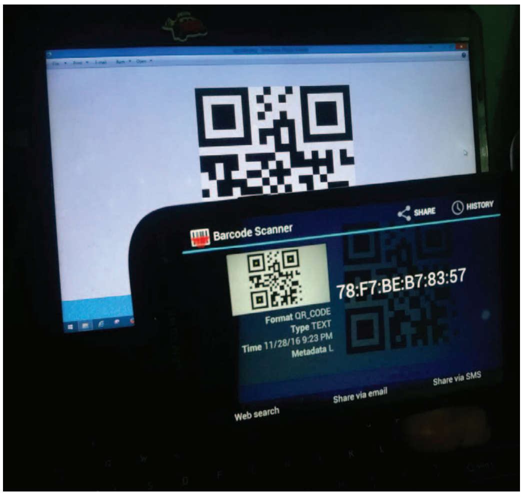

Fig. 6. Bar Code generation procedure for users

V. Experimental Results

Initially, we implemented RSA Algorithm, Genetic Algorithm, and QR Code Generation technique for ensuring and enhancing higher security for cloud computing. The performance of the model depends on the values of encryption and decryption cycle for each approach in terms of accuracy of the packages (kilobyte) transfer time. TABLE I show the time contracted for both encryption and decryption cycle with their average values for each proposed technique.

TABLE I. ACCURACY OF DIFFERENT ALGORITHMS

Steps	RSA Algorithm (ms)		Genetic Algorithm (ms)		QR Code (ms)		Average (%)
Encryption Cycle	84	75	98	92	34	31	**84**
Decryption Cycle	78	68	96	82	22	20	**88**
KB Transfer	768		928		104		

By analyzing the TABLE I, taken time by the Genetic Algorithm (GA) for both encryption and decryption cycle process indicates much upper comparing with the time taken by RSA algorithm and QR code.

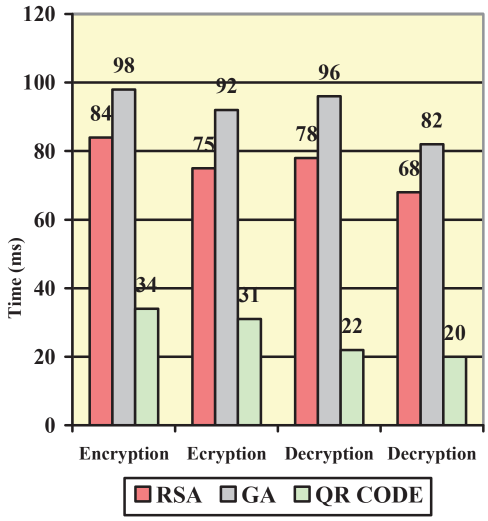

Fig. 7. The graphical representation of time required for encryption and decryption

Fig.7. Shows the graphical representation of time required for both encryption and decryption cycle. For each algorithm calculation of time limit and complexity have been measured in the form of Milliseconds (ms). Here from Fig.7 is visibly shown that in the first attempt, the encryption method of RSA takes 84ms, whereas in a second attempt it takes 75ms. On the Contrary, Genetic Algorithm takes 98ms and 96ms respectively, for successful encryption and QR Code takes 34ms and 31ms respectively. Again in the segment of the decryption process 78ms and 68ms is the taken time for RSA, whereas the GA and QR take (96ms, 82ms) and (22ms, 20ms) correspondingly for decrypting data effectively.

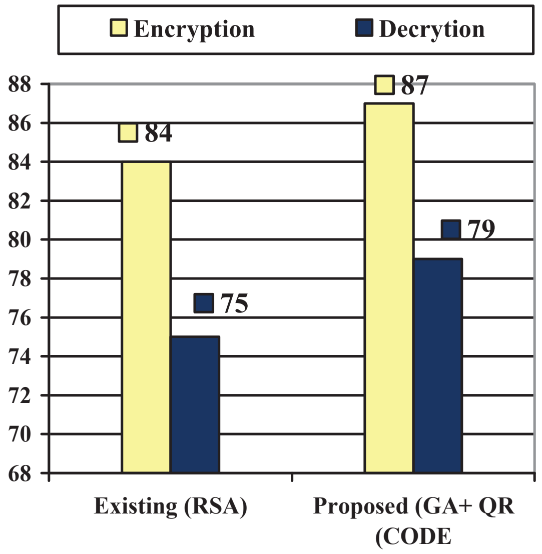

Fig. 8. Comparison of proposed methods with existing algorithm.

Fig.8. Exhibit the statistical comparison among proposed (GA+QR CODE) technique with traditional existing RSA Algorithm. Here our proposed method shows the higher accuracy of 87% in comparison with RSA, which shows 84% of accuracy in terms of Encryption Cycle. Also for decryption cycle Fig.8 represents the higher accuracy of 79% in comparing with the existing method of RSA which shows only 75% of accuracy. After a complete analysis of both Fig.7 and Fig.8 it can be summarized that GA takes longer time compared with RSA whereas QR shows a minor time requirement for encryption and decryption Cycle. Though GA takes longer time than RSA, still our proposed method which is a combination of (GA+QR Code) provide a higher security mechanism for cloud computing devices.

VI. Conclusion

In this paper, we have demonstrated an advance on how to enhance the security of cloud computing with the combination of Genetic Algorithm and QR Code. We have proposed a two-layer security mechanism which consists of Genetic Algorithm for Face Recognition and Quick Response (QR) technique for verification purpose. Currently, existing RSA approach is widely applied for the authentication purpose of cloud computing, devices which consist of the only single layer, but our proposed scheme will offer the security for the second level also. Although our results are not yet good enough to be adopted in any commercial product for cloud computing devices, our approach still has significant prospective and advantages for further development. By comparing the outcomes with existing we have demonstrated the potency of using the combination of both GA and QR. In this context of enhancing the security of cloud computing using our suggested approach with existing one also holds potential. So, we believe that the outcome of our proposed model of providing a 2-Layer Security mechanism where the combination of GA and RA has been shown with Existing RSA model certainly ensure the superior security of cloud server. In the near future, we will likewise try to implement Biometric Approach and evaluate the performance of whole approached model in terms of security enhancement of cloud computing.

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